Proof of Lemma 0.1.3.

Let $P^* \supseteq P$ be a refinement of P. Since partition contains only finite elements, WLOG, assume that $P^* = P \cup \{s\}$ s.t. $s \notin P = \{x_0, \cdots, x_n\}$

Then, for some $1 \le j \le n$ $x_{j-1} < s < x_j$. Now,

$$\begin{aligned} \mathcal{U}(f,P,\alpha) - \mathcal{U}(f,P^*,\alpha) &= \sup_{t\in[x_{j-1},x_j]} f(t)\cdot[\alpha(x_j)-\alpha(s)] + \sup_{t\in[x_{j-1},x_j]} f(t)\cdot[\alpha(s)-\alpha(x_{j-1})] \\ &\quad - \sup_{t\in[x_{j-1},s]} f(t)\cdot[\alpha(x_j)-\alpha(s)] - \sup_{t\in[s,x_j]} f(t)\cdot[\alpha(s)-\alpha(x_{j-1})] \\ &\ge 0. \end{aligned}$$

▨

Proof of Lemma 0.1.4. X Fail try.

Since $f, g \in \mathcal{R}(\alpha)$, $\overline{\int_a^b} f\, d\alpha = \underline{\int_a^b} f\, d\alpha$ and $\overline{\int_a^b} g\, d\alpha = \underline{\int_a^b} g\, d\alpha$. Now,

$$\begin{aligned} \overline{\int_a^b} f+g\, d\alpha &= \inf_{P\in\mathcal{P}[a,b]} \mathcal{U}(f+g,P,\alpha) = \inf_{P\in\mathcal{P}[a,b]} \left[\sum_{i=1}^n M_i(f+g)\cdot\Delta\alpha_i\right] \\ &\le \inf_{P\in\mathcal{P}[a,b]} \left[\sum_{i=1}^n M_i(f)\cdot\Delta\alpha_i + \sum_{i=1}^n M_i(g)\cdot\Delta\alpha_i\right] \end{aligned}$$

Proof of Thm 0.1.4

1) $\Rightarrow$ 2). Since $f \in \mathcal{R}(\alpha)$, given $\varepsilon > 0$, there exists a Partition $P \in \mathcal{P}[a,b]$ s.t

$$\int_a^b f\, d\alpha \le U(f,P,\alpha) < \int_a^b f\, d\alpha + \frac{\varepsilon}{2} \quad \text{and} \quad \int_a^b f\, d\alpha \ge L(f,P,\alpha) > \int_a^b f\, d\alpha - \frac{\varepsilon}{2}$$

Ask the next step to a toddler on the street.

2) $\Rightarrow$ 1). It is clear, because for any partition $P \in \mathcal{P}[a,b]$,

$$\overline{\int_a^b} f\, d\alpha - \underline{\int_a^b} f\, d\alpha \le U(f,P,\alpha) - L(f,P,\alpha)$$

To show 1) $\Leftrightarrow$ 3), we prove a statement: If 2) holds, then for any $1 \le i \le n$,

for arbitrary $s_i, t_i \in [x_{i-1}, x_i]$, $\sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i < \varepsilon$.

Since $|f(s_i) - f(t_i)| \le M_i - m_i$,

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i \le U(f,P,\alpha) - L(f,P,\alpha) < \varepsilon.$$

1) $\Rightarrow$ 3). Suppose $f \in \mathcal{R}(\alpha)$. Then, by 2), there exists a Partition $P_0 \in \mathcal{P}[a,b]$ s.t

$$U(f,P_0,\alpha) - L(f,P_0,\alpha) < \varepsilon$$

Now, for any refinement $P \supset P_0$,

$$U(f,P,\alpha) - L(f,P,\alpha) \le U(f,P_0,\alpha) - L(f,P_0,\alpha) < \varepsilon$$

Clearly, regardless of choice of t_i,

$$L(f,P,\alpha) \le S(f,P,\alpha) \le U(f,P,\alpha)$$

and

$$U(f,P,\alpha) \ge \int f\, d\alpha > L(f,P,\alpha)$$

Therefore, $\left| S(f,P,\alpha) - \int f\, d\alpha \right| \le U(f,P,\alpha) - L(f,P,\alpha) < \varepsilon.$

3) $\Rightarrow$ 1) is trivial.

Proof of lemma 0.1.5.

Let $f, g \in \mathcal{R}(\alpha)$ be given. Given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a,b]$ such that

$$\mathcal{U}(f,P,\alpha) - \mathcal{L}(f,P,\alpha) < \frac{\varepsilon}{2} \text{ and } \mathcal{U}(g,P,\alpha) - \mathcal{L}(g,P,\alpha) < \frac{\varepsilon}{2}$$

(Technically, take $P_1, P_2 \in \mathcal{P}[a,b]$ for f, g respectively, and put $P = P_1 \cup P_2$) Now,

$$\mathcal{U}(f+g,P,\alpha) - \mathcal{L}(f+g,P,\alpha) \leq \mathcal{U}(f,P,\alpha) + \mathcal{U}(g,P,\alpha) - \mathcal{L}(f,P,\alpha) - \mathcal{L}(g,P,\alpha) < \varepsilon.$$

And the equality is obtained by; given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a,b]$ s.t

$$\begin{aligned} \int_a^b f\,d\alpha + \int_a^b g\,d\alpha - \varepsilon &\leq \mathcal{L}(f,P,\alpha) + \mathcal{L}(g,P,\alpha) \leq \mathcal{L}(f+g,P,\alpha) \\ &\leq \int_a^b f+g\,d\alpha \\ &\leq \mathcal{U}(f+g,P,\alpha) \leq \mathcal{U}(f,P,\alpha) + \mathcal{U}(g,P,\alpha) \leq \int_a^b f\,d\alpha + \int_a^b g\,d\alpha + \varepsilon \end{aligned}$$

The second assertion is clear. Now, we can say that

$\mathcal{R}(\alpha)$ is a $\mathbb{R}$-vector space.

Proof of lemma 0.1.6.

Suppose that $f \in \mathcal{R}(\alpha)$ on $[a,b]$. Given $\varepsilon > 0$, there exists $P \in \mathcal{P}[a,b]$ s.t

$$\mathcal{U}(f,P,\alpha) - \mathcal{L}(f,P,\alpha) < \varepsilon.$$

Put $P^* = P \cup \{c\}$. Then, for $P_1 = P^* \cap [a,c]$ and $P_2 = P^* \cap [c,b]$. Now,

$$\begin{aligned} \overline{\int_a^c} f\,d\alpha - \underline{\int_a^c} f\,d\alpha + \overline{\int_c^b} f\,d\alpha - \underline{\int_c^b} f\,d\alpha &\leq \mathcal{U}(f,P_1,\alpha) - \mathcal{L}(f,P_1,\alpha) + \mathcal{U}(f,P_2,\alpha) - \mathcal{L}(f,P_2,\alpha) \\ &= \mathcal{U}(f,P^*,\alpha) - \mathcal{L}(f,P^*,\alpha) < \varepsilon. \end{aligned}$$

Similarly, the converse holds.

「Proof of Thm 0.1.8.
Given $\varepsilon>0$, take $\eta>0$ s.t $[\alpha(b)-\alpha(a)]\cdot\eta<\varepsilon$.
Since f is Continuous on the Compact set $[a,b]$, it is Continuous uniformly.
Thus, there exists $\delta>0$ s.t

$$\forall x,y\in[a,b],\quad |x-y|<\delta \Rightarrow |f(x)-f(y)|<\eta.$$

Now, let $P\in\mathcal{P}[a,b]$ be a partition such that for all i, $\Delta x_i<\delta$ ($P=\{x_0,\cdots,x_n\}$)
Then, for all i, $M_i-m_i\le\eta$. Now,

$$U(f,P,\alpha)-L(f,P,\alpha)=\sum_{i=1}^{n}(M_i-m_i)\cdot\Delta\alpha_i\le\sum_{i=1}^{n}\eta\cdot\Delta\alpha_i=\eta\cdot\sum_{i=1}^{n}\Delta\alpha_i=\eta\cdot[\alpha(b)-\alpha(a)]<\varepsilon.$$

「Proof of Thm 0.1.9. Alternatively, since monotone function can have only Countively simple disconitinuity, if no such shared discontinuities with f and α, then countively discontinuous f is R.S. integrable.
But, in this, prove by elementary, using properties of monotone.

Since α is Continuous, for each $n\in\mathbb{N}$, there exists a partition $P_n\in\mathcal{P}[a,b]$,

$$\Delta\alpha_i=\frac{\alpha(b)-\alpha(a)}{n}\ (\overset{\text{def}}{=}\Delta\alpha)\quad\text{for all } i$$

(More specifically, let $n\in\mathbb{N}$ be fixed. By the Intermidiate Value Theorem,
there exists $x_1\in[a,b]$ s.t $\alpha(x_1)=\alpha(a)+\dfrac{\alpha(b)-\alpha(a)}{n}$. Inductively,

there exists $x_2\in[x_1,b]$ s.t $\alpha(x_2)=\alpha(a)+2\cdot\dfrac{\alpha(b)-\alpha(a)}{n}$

$$\vdots$$

$x_{n-1}\in[x_{n-2},b]$ s.t $\alpha(x_{n-1})=\alpha(a)+(n-1)\cdot\dfrac{\alpha(b)-\alpha(a)}{n}$

And take $x_n=b$. Essentially, the Existence such partition is obtained by I.V.P.
But, Monotone can have only first-kind discontinuity and
IVP implies it can not have first-kind discontinuity, thus Monotone IVT $\Rightarrow$ Continuous)

Given $\varepsilon>0$, WLOG, Assume f is monotone increasing. Take $n\in\mathbb{N}$ s.t $\dfrac{[\alpha(b)-\alpha(a)]\cdot[f(b)-f(a)]}{n}<\varepsilon$.

$$U(f,P_n,\alpha)-L(f,P_n,\alpha)=\sum_{i=1}^{n}[M_i-m_i]\cdot\Delta\alpha_i=\sum_{i=1}^{n}[f(x_i)-f(x_{i-1})]\Delta\alpha_i=\Delta\alpha\cdot\sum_{i=1}^{n}[f(x_i)-f(x_{i-1})]$$

$$=\Delta\alpha\cdot[f(b)-f(a)]<\varepsilon.\quad\blacksquare$$

Proof of Thm. 0.1.10.

Since ϕ is continuous on the compact set $[m, M]$, it is continuous uniformly.

Given $\varepsilon > 0$, there exists $0 < \delta < \varepsilon$ s.t. $|s-t| < \delta \Rightarrow |\phi(s) - \phi(t)| < \varepsilon$.

Since $f \in \mathcal{R}(\alpha)$ on $[a,b]$, there is a partition $\{x_0, \cdots, x_n\}$ on $[a,b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \delta^2$$

Let $M_i = \sup f(t)$ and $m_i = \inf f(t)$ and $M_i^* = \sup (\phi \circ f)(t)$ and $m_i^* = \inf (\phi \circ f)(t)$ where the sup, inf values are taken in $t \in [x_{i-1}, x_i]$.

Divides the index set $\{1, \cdots, n\}$ as;

$$A = \{ i \in \{1, \cdots, n\} \mid M_i - m_i < \delta \} \text{ and } B = \{ i \in \{1, \cdots, n\} \mid M_i - m_i \geq \delta \}.$$

Now, if $i \in A$, then $M_i^* - m_i^* = \sup(\phi \circ f)(t) - \inf(\phi \circ f)(t) \leq \varepsilon$

because $\forall s, t \in [x_{i-1}, x_i]$, $|f(s) - f(t)| \leq M_i - m_i \Rightarrow |\phi(f(s)) - \phi(f(t))| < \varepsilon$.

And, if $i \in B$, then $M_i^* - m_i^* \leq 2 \cdot \sup_{t \in [a,b]} |\phi(t)|$

We obtain that

$$\delta \cdot \sum_{i \in B} \Delta\alpha_i \leq \sum_{i \in B} [M_i - m_i] \cdot \Delta\alpha_i \leq \sum_{i=1}^{n} [M_i - m_i] \cdot \Delta\alpha_i = U(f, P, \alpha) - L(f, P, \alpha) < \delta^2.$$

That is $\sum_{i \in B} \Delta\alpha_i < \delta$. Finally,

$$\begin{aligned}
\sum_{i=1}^{n} [M_i^* - m_i^*] \Delta\alpha_i &= \sum_{i \in A} [M_i^* - m_i^*] \Delta\alpha_i + \sum_{i \in B} [M_i^* - m_i^*] \cdot \Delta\alpha_i \\
&\leq \varepsilon \cdot \sum_{i \in A} \Delta\alpha_i + 2 \cdot \sup_{t \in [m, M]} |\phi(t)| \cdot \sum_{i \in B} \Delta\alpha_i \\
&\leq \varepsilon \cdot [\alpha(b) - \alpha(a)] + 2 \sup |\phi| \cdot \delta \\
&\leq \varepsilon \cdot [\alpha(b) - \alpha(a) + 2 \sup |\phi|]
\end{aligned}$$

Proof of Thm 0.1.11.

Suppose that $f \in \mathcal{R}(\alpha)$. Given $\varepsilon > 0$, there is a partition $P = \{x_0, \cdots, x_n\}$ on $[a,b]$ s.t.

$$U(\alpha', P) - L(\alpha', P) < \varepsilon,$$ since α' is continuous, thus $\alpha' \in \mathcal{R}$.

By the Mean-Value Theorem, for each $1 \le i \le n$

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \alpha'(t_i)(x_i - x_{i-1}) = \alpha'(t_i) \cdot \Delta x_i \text{ for some } t_i \in [x_{i-1}, x_i]$$

For any $s_i \in [x_{i-1}, x_i]$,

$$\sum_{i=1}^{n} |\alpha'(s_i) - \alpha'(t_i)| \cdot \Delta x_i \le \sum_{i=1}^{n} [M_i - m_i] \cdot \Delta x_i < \varepsilon.$$

Now,

$$\left| \sum_{i=1}^{n} f(s_i) \cdot \Delta\alpha_i - \sum_{i=1}^{n} f(s_i) \cdot \alpha'(s_i) \cdot \Delta x_i \right|$$

$$= \left| \sum_{i=1}^{n} f(s_i) \cdot \alpha'(t_i) \cdot \Delta x_i - \sum_{i=1}^{n} f(s_i) \alpha'(s_i) \cdot \Delta x_i \right|$$

$$= \left| \sum_{i=1}^{n} [\alpha'(t_i) - \alpha'(s_i)] \cdot f(s_i) \cdot \Delta x_i \right| \le \sup|f| \cdot \varepsilon \quad (*)$$

Therefore, $\left| \overline{\int_a^b} f \, d\alpha - \overline{\int_a^b} f\alpha' \, dx \right| \le |U(f, P, \alpha) - U(f\alpha', P, x)| \le \sup|f| \cdot \varepsilon$

since $(*)$ holds for any refinement of P.

⌈Proof of Thm 0.1.13.

Enough to show 1) <=> 2). Let $\int_a^b f = A$.

Given $\varepsilon > 0$, there exists $P_0 = \{x_0, x_1, \cdots, x_n\}$ s.t $\mathcal{U}(f, P_0) < A + \varepsilon$.

Put $M = \sup|f|$, $\delta_1 = \frac{\varepsilon}{nM}$. Let $P \in \mathcal{P}[a,b]$ s.t $\|P\| < \delta_1$. Denote $P = \{y_0, \cdots, y_m\}$.

Let $I = \{i \in \{1, \cdots, m\} \mid (y_{i-1}, y_i) \cap P_0 \neq \emptyset\}$ (WLOG, $f > 0$. ($f = f + \min f$))

$J = \{j \in \{1, \cdots, m\} \mid (y_{j-1}, y_j) \cap P_0 = \emptyset\}$

Now, $\mathcal{U}(f,P) = \sum_{i=1}^{m} M_i (y_i - y_{i-1})$

$= \sum_{i \in J} M_i (y_i - y_{i-1}) + \sum_{i \in I} M_i (y_i - y_{i-1})$

$< nM\|P\| + \mathcal{U}(f, P_0) < A + 2\varepsilon.$

겹치는게 많아봐야 n개! 더 큰 구간에서 sup!

Similarly $\mathcal{L}(f,P) > A - 2\varepsilon$, for some $\delta_2 > 0$, $\|P\| < \delta_2$ ⊠

⌈Thm. Let $f : [a,b] \to \mathbb{R}$ be a bounded map and continuous except finitely many points. Then f is Riemann integrable.

Proof. Define $P_n = [a,b] \cap \{\frac{k}{2^n} \mid k \in \mathbb{Z}\}$. Let A be a set of discontinuous point of f.

Put N be a number of elements of A. Now, for any $n \in \mathbb{N}$, there is at most $2N$ for $i \in \mathbb{N}$

s.t $A \cap [x_{i-1}, x_i] \neq \emptyset$

where $x_i \in P_n$. Given $\varepsilon > 0$, take $n \in \mathbb{N}$ s.t $\sup|f| \cdot 2N \cdot \frac{1}{2^n} < \frac{\varepsilon}{2}$.

Meanwhile, f is continuous on the set

$$C = \bigcup_{A \cap [x_{i-1}, x_i] = \emptyset} [x_{i-1}, x_i]$$

Therefore there is a partition P' on C s.t $\mathcal{U}(f, P') - \mathcal{L}(f, P') < \frac{\varepsilon}{2}$.

Finally, for $P = P_n \cup P'$,

$\mathcal{U}(f,P) - \mathcal{L}(f,P) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon.$ ⊠

Proof of Lemma 0.1.25.

Let $P_n = \{a, s, s+\frac{1}{n}, b\}$ for large enough $n \in \mathbb{N}$ $(\frac{1}{n} < b-s)$.

Then, $\Delta\alpha_1 = \alpha(s) - \alpha(a) = 0$, $\Delta\alpha_2 = 1$, $\Delta\alpha_3 = 0$. Now

$$\inf_{t\in[s,s+\frac{1}{n}]} f(t) \le \underline{\int} f\,d\alpha \le \overline{\int} f\,d\alpha \le \sup_{t\in[s,s+\frac{1}{n}]} f(t)$$

And since f is continuous at s, $\lim_{n\to\infty} \inf_{t\in[s,s+\frac{1}{n}]} f(t) = \lim_{n\to\infty} \sup_{t\in[s,s+\frac{1}{n}]} f(t) = f(s)$.

Proof of Thm 0.1.26.

Given $\varepsilon>0$, take $N \in \mathbb{N}$ s.t. $\sum_{n=N+1}^{\infty} C_n < \infty$.

Let $\alpha_1(x) = \sum_{n=1}^{N} C_n I(x-s_n)$ and $\alpha_2(x) = \sum_{n=N+1}^{\infty} C_n I(x-s_n)$. Then $\alpha = \alpha_1 + \alpha_2$.

First, $\int_a^b f\,d\alpha_1 = \sum_{n=1}^{N} C_n \cdot f(s_n)$, and next, for $M = \sup|f|$

$$\left|\int_a^b f\,d\alpha_2\right| \le \int_a^b |f|\,d\alpha_2 \le M\cdot\int_a^b 1\,d\alpha_2 = M\cdot\sum_{n=N+1}^{\infty} C_n < M\varepsilon.$$

Now, $\int_a^b f\,d\alpha = \int_a^b f\,d\alpha_1 + \int_a^b f\,d\alpha_2 = \sum_{n=1}^{N} C_n f(s_n) + \int_a^b f\,d\alpha_2$

$\Rightarrow \left|\int_a^b f\,d\alpha - \sum_{n=1}^{N} C_n f(s_n)\right| < M\varepsilon$, and letting $N\to\infty$.

Exercise 6

#1. Suppose α increase on $[a,b]$, and Continuous at $x_0 \in [a,b]$.
If $f(x_0)=1$ and $f(x)=0$ for all $x \neq x_0$, then $f \in \mathcal{R}(\alpha)$ and $\int f\, d\alpha = 0$.

Proof. Let $P_n := \{a, x_0 - \frac{1}{n}, x_0, x_0 + \frac{1}{n}, b\} \cap [a,b]$.

Given $\varepsilon > 0$, there exists $\delta > 0$ s.t. $|x_0 - t| < \delta \Rightarrow |\alpha(x_0) - \alpha(t)| < \frac{\varepsilon}{2}$
Now, for $N \in \mathbb{N}$ s.t. $1 < N\delta$, now,

$$0 = L(f, P_N, \alpha) \leq \underline{\int} f\, d\alpha \leq \overline{\int} f\, d\alpha \leq U(f, P_N, \alpha) < \varepsilon.$$

#2. Suppose that for some $x_0 \in [a,b]$, $f(x_0) > 0$.
Then, there exists $\delta > 0$ s.t. $|t - x_0| < \delta \Rightarrow |f(t) - f(x_0)| < \frac{1}{2} \cdot f(x_0)$.
By the triangle inequality $-\frac{1}{2} f(x_0) + |f(x_0)| < f(t) < \frac{1}{2} f(x_0) + |f(x_0)|$, if $|t - x_0| < \delta$.

Let $P \in \mathcal{P}[a,b]$ be any Partition. Then, for a refinement

$$P^* = P \cup (\{x_0 - \delta, x_0 + \delta\} \cap [a,b]),$$

Now, $\frac{f(x_0)}{2} \cdot \delta \leq U(f, P^*) \leq U(f, P)$

But, we can take $P' \in \mathcal{P}[a,b]$ s.t. $U(f, P') < \frac{f(x_0)}{2} \cdot \delta$ since $\int_a^b f = 0$,
it is Contradiction.

#3. Define $\beta_1, \beta_2, \beta_2$ on $[-1,1]$ s.t

$$\beta_j(x)=0 \text{ if } x<0, \quad \beta_j(x)=1 \text{ if } x>0, \quad \text{and } \beta_1(0)=0, \beta_2(0)=1, \beta_3(0)=\frac{1}{2}.$$

Let f be a bounded map on $[-1,1]$.

a). $f \in \mathcal{R}(\beta_1)$ if and only if $f(0+)=f(0)$ and that then $\int f\, d\beta_1 = f(0)$

b).

c.) $f \in \mathcal{R}(\beta_3)$ if and only if f is continuous at 0.

d). If f is continuous at $x=0$, then

$$\int f\, d\beta_1 = \int f\, d\beta_2 = \int f\, d\beta_3 = 0.$$

Proof. Let $P_n = \{-1, 0, \frac{1}{n}, 1\}$. Then, for each $i=1,2,3$,

$$U(f, P_n, \beta_i) = M_1 \cdot [\beta_i(0) - \beta_i(-1)] + M_2 \cdot [\beta_i(\tfrac{1}{n}) - \beta_i(0)] + M_3 \cdot [\beta_i(1) - \beta_i(\tfrac{1}{n})]$$

$$= M_1 \beta_i(0) + M_2[1-\beta_i(0)] + 0$$

$$= (M_1 - M_2)\beta_i(0) + M_2. \quad \text{Similarly}$$

$$L(f, P_n, \beta_i) = (m_1 - m_2)\beta_i(0) + m_2.$$

When $i=1$, $\beta_1(0)=0$, therefore

$$U(f, P_n, \beta_1) - L(f, P_n, \beta_1) = M_2 - m_2 = \sup_{t\in[0,\frac{1}{n}]} f(t) - \inf_{t\in[0,\frac{1}{n}]} f(t) \to 0 \quad \text{as } n\to\infty,$$

since $f(0+)=f(0)$. To show b), take $P_n = \{-1, -\frac{1}{n}, 0, \frac{1}{n}, 1\}$. Then,

$$U(f, P_n, \beta_i) = M_1 [\beta_i(-\tfrac{1}{n}) - \beta_i(-1)] + M_2 \cdot [\beta_i(0) - \beta_i(-\tfrac{1}{n})] + M_3[\beta_i(\tfrac{1}{n}) - \beta_i(0)] + M_4[\beta_i(1) - \beta_i(\tfrac{1}{n})]$$

$$= M_2 \cdot \beta_i(0) + M_3(1-\beta_i(0)). \quad \text{Similarly}$$

$$L(f, P_n, \beta_i) = m_2 \beta_i(0) + m_3(1-\beta_i(0))$$

When $i=2$, $\beta_2(0)=1$. Therefore

$$U(f, P_n, \beta_2) - L(f, P_n, \beta_2) = M_2 - m_2 \to 0 \quad \text{if } f(0^-)=f(0).$$

when $i=3$, $\beta_3(0)=\frac{1}{2}$, thus

$$U(f, P_n, \beta_3) - L(f, P_n, \beta_3) = \frac{1}{2}[M_2 - m_2 + M_3 - m_3] \to 0 \quad \text{if } f(0+)=f(0)=f(0+).$$

Moreover, $U(f, P_n, \beta_i) \to f(0)\cdot[\beta_i(0) + 1 - \beta_i(0)] = f(0)$ as $n\to\infty$. $\boxtimes$

「#4. Dirichlet's map is <u>Not</u> Riemann-integrable.

Proof. Let $P \in \mathcal{P}[a,b]$ be any partition. Then, by the density of Rationals and irrationals,

$M_i = 1$, $m_i = 0$ for all i. That is,

$$\mathcal{U}(f,P) = 1 \quad \text{and} \quad \mathcal{L}(f,P) = 0.$$

Thus, $P \notin \mathcal{R}(x)$. (WLOG, $[a,b] := [0,1]$)

「#5. f is bounded, and $f^2 \in \mathcal{R}$, then $f \in \mathcal{R}$? No!

$$f : [0,1] \to \mathbb{R} : x \mapsto \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \notin \mathbb{Q} \end{cases}$$

$f^2 = 1$, thus integrable, but f is not.

Meanwhile, if $f^3 \in \mathcal{R}$, then $f \in \mathcal{R}$ because the real value of $\sqrt[3]{f^3}$ is unique.

⌈#6. Let C be a Cantor set.

Define the Charateristic fuction of A as

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Clearly, $\chi_A + \chi_{A^c} = 1$. Now, observe that for each $k \in \mathbb{N}$ and any $P \in \mathcal{P}[a,b]$,

$$\mathcal{U}(f,P) - \mathcal{L}(f,P) \le \mathcal{U}(f\cdot\chi_{C_k},P) + \mathcal{U}(f\cdot\chi_{C_k^c},P) - \mathcal{L}(f\cdot\chi_{C_k},P) - \mathcal{L}(f\cdot\chi_{C_k^c},P)$$

Since $C \subseteq C_k$, $C_k^c \subseteq C^c$. Therefore, $f\cdot\chi_{C_k^c}$ is continuous map except finitely many points, thus Riemann-integrable. Meanwhile, for each $m \in \mathbb{N}$, define

$$P_m = \left\{ \frac{a}{3^m} \,\middle|\, 0 \le a \le 3^m,\ a \in \mathbb{Z} \right\}$$

Then, $\mathcal{U}(f\cdot\chi_{C_k},P_m) - \mathcal{L}(f\cdot\chi_{C_k},P_m) \le 2\cdot\sup|f|\cdot 2^k\cdot[3^{m-k}+2]\cdot\frac{1}{3^m} \to 0$ as $k,m \to \infty$

Thus proved. (More Technically, we can take $m = k+1$.)

⌈Thm. Let $f:[a,b] \to \mathbb{R}$ be a bounded map and continuous except countably many points. Then f is Riemann-integrable.

Proof. Define $P_n = [a,b] \cap \left\{ \frac{k}{2^n} \,\middle|\, k \in \mathbb{Z} \right\}$. Let A be a set of discontinuous point of f.

Denote $A = \{ d_n \mid n \in \mathbb{N} \}$, $d_n < d_{n+1}$. Observe that for $n \ge 1$,

$\mathcal{U}(f,P_n) \le$

#7. Let $f \in \mathcal{R}$ on $(0,1]$, and $f \in \mathcal{R}$ on $[c,1]$ for any $c>0$.

Define
$$\int_0^1 f(x)\,dx = \lim_{c \to 0} \int_c^1 f(x)\,dx$$
if the limit exists.

a). If $f \in \mathcal{R}$ on $[0,1]$, then this definition equals to old one.

b). Construct f s.t the limit exists, but $|f|$ does not.

Proof.

#8. Suppose $f \in \mathcal{R}$ on $[a,b]$ for every $b>a$ where a is fixed. Define
$$\int_a^\infty f(x)\,dx = \lim_{b \to \infty} \int_a^b f(x)\,dx$$
if the limit exists. Assume $f(x) \geq 0$ and decreases monotonically on $[1,\infty)$. Prove
$\int_1^\infty f(x)\,dx$ converges if and only if $\sum_{n=1}^{\infty} f(n)$ converges.

Proof.

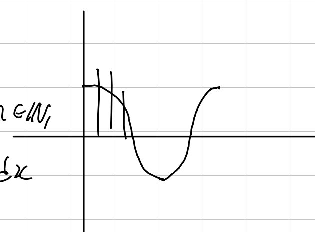
#9. Let $b>0$. Then,

$$\int_0^b \frac{\cos x}{1+x}\,dx = \frac{\sin x}{1+x}\Big|_0^b + \int_0^b \frac{\sin x}{(1+x)^2}\,dx$$
$$= \frac{\sin b}{1+b} + \int_0^b \frac{\sin x}{(1+x)^2}\,dx.$$

Since $\int_0^b \left|\frac{\sin x}{(1+x)^2}\right| dx \le \int_0^b \frac{1}{(1+x)^2}\,dx = \frac{-1}{1+x}\Big|_0^b = 1 - \frac{1}{1+b} \to 1$ as $b\to\infty$,

$\int_0^\infty \frac{\sin x}{(1+x)^2}\,dx$ exists, so is $\int_0^\infty \frac{\cos x}{1+x}\,dx$.

But $\int_0^\infty \frac{|\cos x|}{1+x}\,dx$ does not exists, because



In $n\pi + [-\frac{\pi}{3}, \frac{\pi}{3}]$, $|\cos x| \ge \frac{1}{2}$. Now, define for each $n\in\mathbb{N}$,

$I_n = \int_{-\frac{\pi}{3}+n\pi}^{\frac{\pi}{3}+n\pi} \frac{|\cos x|}{1+x}\,dx$. Clearly, $\sum_{n=1}^{m} I_n \le \int_0^{\frac{\pi}{3}+m\pi} \frac{|\cos x|}{1+x}\,dx$

But $I_n \ge \frac{2\pi}{3}\cdot\frac{1}{2}\cdot \int_{-\frac{\pi}{3}+n\pi}^{\frac{\pi}{3}+n\pi} \frac{1}{1+x}\,dx$, thus proved.

(Since $\frac{1}{1+x}$ decreasing, $\sum_{n=1}^{\infty} \int_{-\frac{\pi}{3}+n\pi}^{\frac{\pi}{3}+n\pi} \frac{1}{1+x} \ge 2\int_0^\infty \frac{1}{1+x}$.) $\square$

#12. Let $f : [a,b] \to \mathbb{R}$ be given with $f \in \mathcal{R}(\alpha)$.

For any $\varepsilon > 0$, there exists a continuous $g : [a,b] \to \mathbb{R}$ s.t. $\|f - g\|_2 < \varepsilon$.

Proof. Given $\varepsilon > 0$, take $P = \{x_0, \cdots, x_n\} \in \mathcal{P}[a,b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

Define $g(t) := \dfrac{x_i - t}{x_i - x_{i-1}} \cdot f(x_{i-1}) + \dfrac{t - x_{i-1}}{x_i - x_{i-1}} \cdot f(x_i) \quad (x_{i-1} \le t \le x_i)$

Then, g is continuous on $[a,b]$: (x_{i-1}, x_i), clearly continuous and

$g(x_i^-) = g(x_i) = g(x_i^+)$

Moreover, $|g(t) - f(t)| \le M_i - m_i$ for any $1 \le i \le n$, therefore

$|g(t) - f(t)| \cdot \Delta\alpha_i \le [M_i - m_i] \cdot \Delta\alpha_i$.

$$\|g - f\|_2 = \left[\sum_{i=1}^{n} \int_{x_{i-1}}^{x_i} |g - f|^2 \, d\alpha \right]^{\frac{1}{2}}$$

#13. Define $f(x) = \int_x^{x+1} \sin(t^2) dt$.

a). Prove that for $x>0$, $|f(x)| < \frac{1}{x}$.

b) Prove that $2xf(x) = \cos(x^2) - \cos[(x+1)^2] + r(x)$ where $|r(x)| < \frac{c}{x}$ for some $c \in \mathbb{R}$

c) Evaluate $\limsup_{x\to\infty} xf(x)$ and $\liminf_{x\to\infty} xf(x)$

d). Does $\int_0^\infty \sin(t^2)\, dt$ Converges?

Proof. Put $t^2 = u$. Then $2t\, dt = du$,

$$f(x) = \int_x^{x+1} \sin(t^2)\, dt = \int_{x^2}^{(x+1)^2} \sin u \cdot \frac{1}{2\sqrt{u}}\, du$$

$$= -\frac{\cos u}{2\sqrt{u}}\Big|_{x^2}^{(x+1)^2} - \int_{x^2}^{(x+1)^2} \cos u \cdot \frac{1}{4u^{\frac{3}{2}}}\, du$$

$$= \frac{\cos x^2}{2x} - \frac{\cos(x+1)^2}{2(x+1)} - \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}}\, du$$

Now, $$|f(x)| \le \frac{1}{2x} + \frac{1}{2(x+1)} + \int_{x^2}^{(x+1)^2} \frac{1}{4u^{\frac{3}{2}}}\, du$$

$$= \frac{1}{2x} + \frac{1}{2(x+1)} + \left(-\frac{1}{2}\right)\cdot\left[u^{-\frac{1}{2}}\Big|_{x^2}^{(x+1)^2}\right]$$

$$= \frac{1}{2x} + \frac{1}{2(x+1)} - \frac{1}{2}\left[\frac{1}{x+1} - \frac{1}{x}\right]$$

$$= \frac{1}{x}$$

Meanwhile, $$2xf(x) = \cos x^2 - \cos(x+1)^2 + \cos(x+1)^2 \cdot \frac{1}{2(x+1)} - 2x\cdot\int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}}\, du$$

observe that

$$\left|\cos(x+1)^2 \cdot \frac{1}{2(x+1)} - 2x\cdot\int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}}\, du\right| \le \frac{1}{2}\cdot\left|\frac{1}{x+1}\right| + 2|x|\cdot\int_{x^2}^{(x+1)^2} \frac{1}{4u^{\frac{3}{2}}}\, du$$

$$\le \frac{1}{2|x|} + \frac{1}{|x+1|} \le \frac{3}{2x}.$$

Clearly, $\limsup_{x\to\infty} r(x) = \liminf_{x\to\infty} r(x) = 0$. And,

$\cos x^2 - \cos(x+1)^2 =$

Clearly, $\limsup_{x\to\infty} r(x) = \liminf_{x\to\infty} r(x) = 0$. And, to evaluate upper and lower limit of the remain, take $x_n = 2n\cdot\sqrt{\pi}$ and $y_n = (2n+1)\sqrt{\pi}$. Now,

$$\cos x_n^2 - \cos(x_n+1)^2 = 1 - \cos(2n\sqrt{\pi}+1)$$

$$\cos y_n^2 - \cos(y_n+1)^2 = -1 + \cos([2n+1]\sqrt{\pi}+1)$$

Since $\{\cos(n\alpha) \mid n\in\mathbb{N}\}$ is dense in $[-1,1]$,

$$\limsup_{x\to\infty}[\cos x^2 - \cos(x+1)^2] = 2 \quad \text{and} \quad \liminf_{x\to\infty}[\cos x^2 - \cos(x+1)^2] = -2.$$

Finally, let's determine whether $\int_0^\infty \sin t^2\,dt$ Converges or diverges.

Since $\int_0^\infty \sin t^2\,dt = \int_0^\infty t\cdot\sin t^2\cdot\frac{1}{t}\,dt$

$\sum_{k=0}^{\infty} kf(k)\cdot\frac{1}{k}$ exists,

But, if $\lim_{N\to\infty}\sum_{k=0}^{N-1} f(k)$ exists, that is,

Since $\underline{\int_0^N \sin t^2\,dt = \sum_{k=0}^{N-1}\int_k^{k+1}\sin t^2\,dt} = \underline{\sum_{k=0}^{N-1} f(k)}$

Using the Dirichlet's Test, then

$$\int_0^\infty \sin t^2\,dt = \underline{\sum_{k=0}^{\infty} f(k)} = \underline{\sum_{k=0}^{\infty} kf(k)\cdot\frac{1}{k}}$$

Converges

#15. Suppose f is real, continuously differentiable, $f(a)=f(b)=0$, with

$$\int_a^b (f\circ f)(x)\,dx = 1.$$

Prove that $\int_a^b x f(x) f'(x)\,dx = -\frac{1}{2}$ and $\int_a^b [f'(x)]^2\,dx \cdot \int_a^b x^2 (f\circ f)(x)\,dx > \frac{1}{4}$.

Proof. $$\int_a^b f^2(x)\,dx = x f^2(x)\Big|_a^b - \int_a^b x\cdot 2f(x)f'(x)\,dx = 1$$

Since the first term on the right is zero,

$$\int_a^b x\cdot f(x) f'(x)\,dx = -\frac{1}{2}.$$

Now, $$\int_a^b [f'(x)]^2\,dx \cdot \int_a^b [x f(x)]^2\,dx \geq \left|\int_a^b x f(x) f'(x)\,dx\right|^2 = \frac{1}{4}$$

By Hölder's inequality.

Prop. Let $f:[a,b]\to\mathbb{R}$ be a differentiable map.
If f is bounded variation, then

$$\int_a^b |f'(x)|\,dx \le TV(f)$$

Proof. For any partition $P=\{x_0,\cdots,x_n\}$ on $[a,b]$, the Mean-value Theorem gives that

$$V(f,P)=\sum_{i=1}^n |f(x_i)-f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)|\cdot\Delta x_i \to \int_a^b |f'(x)|\,dx \quad \text{as } \|P\|\to 0.$$

Thus,